Orientation estimates of horizontal components for OBSIC metrics hub

September 3 2024

Method. Rayleigh Wave Arrival Angles [DLOPy; Doran and Laske, 2017]

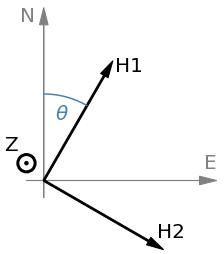
Code. https://github.com/jbrussell/DLOPy_v1.0 **,** https://github.com/kschramm-usgs/DLOPy

Implementation. The default DLOPy input parameters were used. The data window was selected for each site using the spectrograms of the three seismometer channels (HHZ, HH1 and HH2 or BHZ, BH1, BH2).

OBSIC Horizontal Component Orientation Convention

Left-handed coordinate system with positive Z pointing upwards and positive horizontal components H1 and H2 as shown in figure.

θ shows the convention for the DLOPy orientation estimates: the clockwise angle from North to H1, from 0 to 360°.



Results. For each experiment, in stations table and in downloadable stations .csv file.

Orientation	In degrees, as defined above and by Doran and Laske [2017].
Error	In degrees, as defined above and by Doran and Laske [2017].

Limitations. These are semi-automated results using a single method with default input parameters. Other methods may yield more precise or more accurate results, especially for sites with high noise or low data quality.

*Doran, A. K., and G. Laske, 2017, Ocean-bottom seismometer instrument orientations via automated Rayleigh-wave arrival-angle measurements, Bull. Seis. Soc. Am., **107 (2),** 691-708, doi: 10.1785/0120160165.*